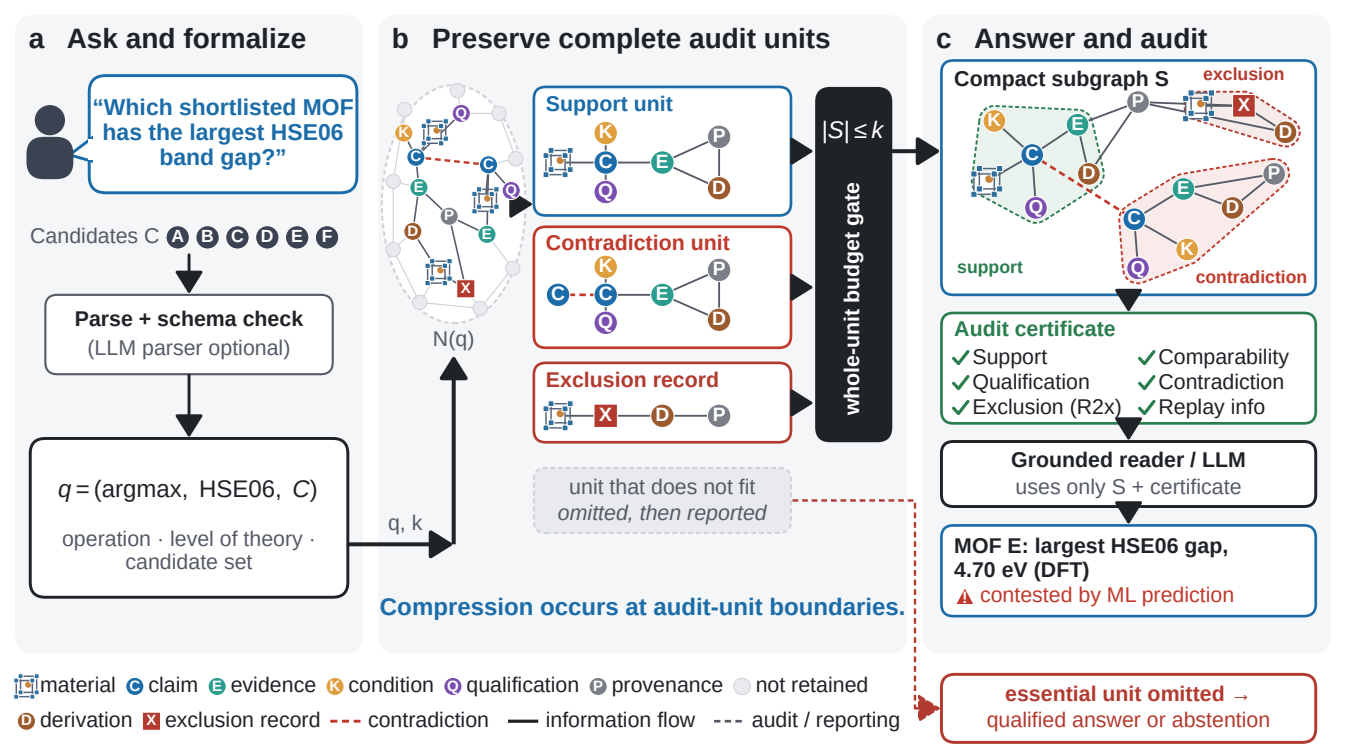

a Ask and formalize
"Which shortlisted MOF has the largest HSE06 band gap?"
Candidates C A B C D E F
Parse + schema check
(LLM parser optional)
q = (argmax, HSE06, C)
operation · level of theory · candidate set
q, k
N(q)
b Preserve complete audit units
Support unit
Contradiction unit
Exclusion record
|S| ≤ k
whole-unit budget gate
unit that does not fit
omitted, then reported
Compression occurs at audit-unit boundaries.
c Answer and audit
Compact subgraph S
exclusion
support
contradiction
Audit certificate
Support
Comparability
Qualification
Contradiction
Exclusion (R2x)
Replay info
Grounded reader / LLM
uses only S + certificate
MOF E: largest HSE06 gap, 4.70 eV (DFT)
contested by ML prediction
material claim evidence condition qualification provenance not retained
derivation exclusion record contradiction information flow audit / reporting
essential unit omitted →
qualified answer or abstention